

DEBSIM: Productive Friction for Learning Transfer in LLM-Based Argumentation Tutoring

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Abstract. Most argumentation tutoring systems evaluate in-system performance rather than learning transfer, and the generative capabilities of large language models (LLMs) risk bypassing the effortful cognitive processes that tutoring aims to develop—a tension known as the assistance dilemma. We present DEBSIM, an LLM-based argumentation tutoring system that addresses this dilemma through productive friction, scaffolding debate preparation across three stages—Toulmin knowledge component structuring, evidence validation with verification friction, and counterargument simulation—while requiring learner effort at cognitively meaningful decision points. In a within-subjects experiment ($N=20$), participants prepared with DEBSIM for one debate topic and without it for another, then engaged in live debates without system access. Participants in the DEBSIM condition produced openings with higher knowledge component completeness and shifted toward evidence-based discourse patterns. These findings offer a transfer-based evaluation methodology for argumentation tutoring, preliminary evidence that productive friction can help address the assistance dilemma in the LLM era, and an inner/outer-loop scaffolding architecture for supporting argumentation skill transfer from AI-assisted preparation to unassisted performance.

Keywords: Argumentation tutoring · Intelligent tutoring systems · Adaptive scaffolding · Assistance dilemma · Learning transfer · Productive friction · LLM-based pedagogical agents

1 Introduction

Argumentation—the ability to construct, support, and defend claims through evidence and reasoning—is a foundational skill across education, professional practice, and civic life [1, 2]. The intelligent tutoring systems (ITS) community has pursued computer-supported argumentation for over three decades, progressing from early visual mapping environments such as Belvedere [19] to graph-based legal reasoning tutors such as LARGO [36] and contemporary NLP-powered

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feedback systems [6, 21]. Despite this sustained effort, two persistent challenges limit existing approaches.

The first challenge concerns evaluation methodology. Most argumentation systems evaluate in-system performance—the quality of arguments produced with tool support—rather than assessing learning transfer [35]. This limitation reflects the distinction between performance (task success with assistance) and learning (task success after assistance is removed) [14, 13]. Without transfer-based evaluation, it remains unclear whether argumentation tools enhance learner competence or merely augment in-task output.

The second challenge concerns LLM-based tutoring design. Large language models can generate complete arguments on demand [3, 4] and produce counterarguments by identifying weak premises [7], making it easy for learners to bypass the cognitive processes tutoring aims to develop. This intensifies what Koedinger and Aleven [37] term the assistance dilemma: how can systems harness LLM capabilities to support—rather than replace—the effortful processing that enables durable learning?

We present DEBSIM, an LLM-based argumentation tutoring system that addresses both challenges. DEBSIM scaffolds debate preparation across three stages—argument structuring using Toulmin knowledge components (KCs) [30], evidence validation with verification friction, and counterargument simulation for adversarial rehearsal—while deliberately requiring learner effort at cognitively meaningful decision points. The system automates mechanical operations (search execution, formatting, structure propagation) while withholding automation for cognitive operations (evidence evaluation, counterargument generation, structural revision). We define mechanical operations as those that do not require domain reasoning—executing a search query, reformatting text, propagating structural edits—as distinct from cognitive operations that require evaluative judgement, such as assessing source credibility or generating rebuttals. We call the resulting design strategy *productive friction*: a design response to the assistance dilemma in the LLM era.

Our evaluation assesses learning transfer: participants prepare using DEBSIM but have no system access during live debate. In a within-subjects study ($N=20$), DEBSIM preparation was associated with higher KC completeness in debate openings—particularly in backing and qualifier components—as well as a shift from adversarial negation toward evidence-based inquiry. Five of twelve self-report measures survived Benjamini–Hochberg correction at FDR $q<.05$.

This research contributes: (1) a transfer-based evaluation methodology that distinguishes learning from supported performance; (2) preliminary evidence that productive friction is associated with transfer of argumentative skills, instantiated through the distinction between mechanical and cognitive automation; and (3) an inner/outer-loop scaffolding architecture in which multi-stage KC construction forms iterative refinement cycles.

2 Related Work

Prior work on argumentation tutoring, LLM-assisted learning, and scaffolding design provides the foundation for DEBSIM. Together, these strands motivate our focus on whether learners can transfer argumentation skills beyond scaffolded preparation.

2.1 Evaluating Argumentation Learning: In-System Output vs. Transfer

Argumentation tutoring systems have progressed through several generations [35], yet most evaluate learner outputs produced with tool support rather than what learners can do after the tool is removed. Early systems such as Belvedere [19], LARGO [36], and the Deliberatorium [20] established foundational patterns for visual mapping, graph-based reasoning, and structured deliberation. More recent systems incorporate adaptive feedback: AL [6], ArgueTutor [21], and ArgumentFeedback [22]. Surveys confirm that detailed adaptive feedback outperforms generic feedback [38, 39, 8].

Most of these systems rely on Toulmin’s argumentation model [30, 26], which decomposes arguments into six components (claim, data, warrant, backing, qualifier, rebuttal). We adopt these six components as knowledge components (KCs) in the KLI framework [13]: each represents an independently scaffoldable and assessable argumentation skill [23]. This mapping is warranted because each Toulmin component involves a distinct cognitive operation—e.g., identifying evidential support (data) requires different reasoning than anticipating conditions of exception (qualifier)—and learners develop proficiency in these components at different rates, satisfying the KLI criterion of decomposable skill units. However, Toulmin suits monological construction better than dynamic debate [31]; assessing interactive argumentation additionally requires functional categories such as challenge, clarification, and concession.

The critical gap is evaluative: the KLI framework distinguishes performance from learning [14, 13], yet nearly all existing systems measure only the former [35]. DEBSIM addresses this gap by assessing transfer from scaffolded preparation to unassisted live debate, using Toulmin coding for openings and Walton’s categories [31] for interactive exchanges.

2.2 LLMs and the Intensified Assistance Dilemma

The assistance dilemma—how learning environments should balance giving and withholding information [37]—has become more acute with LLMs. Chain-of-thought prompting [3, 24] enables structured argument generation, multi-agent architectures [4, 5] can simulate opposing viewpoints, and educational applications have explored LLM-based debate partnerships [12]. At the same time, LLMs cannot maintain consistent beliefs under challenge [11], exhibit systematic biases [25], and show variable argumentation quality [9]. Empirical work further shows that users often append hallucinated LLM outputs to their own

correct answers rather than rejecting them, and that cognitive forcing functions intended to counter this can paradoxically amplify overreliance when model outputs are inaccurate [10].

The deeper concern is pedagogical. When an LLM generates a complete argument, the risk is automation complacency [37]—surface-level engagement that does not survive support removal. DEBSIM responds by distinguishing mechanical operations (search, formatting, structure propagation) that the system automates from cognitive operations (evidence evaluation, counterargument generation, structural revision) that remain with the learner.

2.3 From Desirable Difficulties to Productive Friction

Cognitive scaffolding theory [28, 29] holds that well-designed support enables learners to perform beyond their independent capacity, with competence internalised as scaffolds are removed. The desirable difficulties framework [15] complements this by showing that effortful acquisition conditions—generation rather than recognition [16], retrieval practice rather than re-study [17]—enhance long-term retention and transfer. Recent empirical evidence sharpens this concern for AI-assisted workflows: in a survey of 319 knowledge workers, higher confidence in generative AI was associated with less critical thinking, as users shifted cognitive effort from active problem-solving toward passive verification of AI outputs [18].

Prior argumentation systems such as Belvedere [19], LARGO [36], and AL [6] embedded learner effort through the structural demands of their interfaces; LLM-based tutoring differs in that such effort must be introduced by design rather than inherited from the interface. We use three related terms with distinct scope: *scaffolding* [28] for support structures, *desirable difficulties* [15] for cognitive conditions that enhance durable learning, and *productive friction* for interface-level resistance that operationalises desirable difficulties in LLM-based tutoring. DEBSIM implements a binary transition: full scaffolding during preparation, complete removal during debate.

3 System Design

DEBSIM scaffolds debate preparation through three stages—Argument Formation, Evidence Enhancement, and Counterargument Simulation—each targeting specific Toulmin KCs [30] and embedding productive friction where learners must make active decisions rather than passively accepting AI outputs. We describe the system from two perspectives: the learner’s interaction workflow and the backend architecture.

3.1 Learner Interaction Workflow

Fig. 1 shows the DEBSIM interface. A learner’s session proceeds through three stages with non-linear navigation: learners may revisit any completed stage at any time.

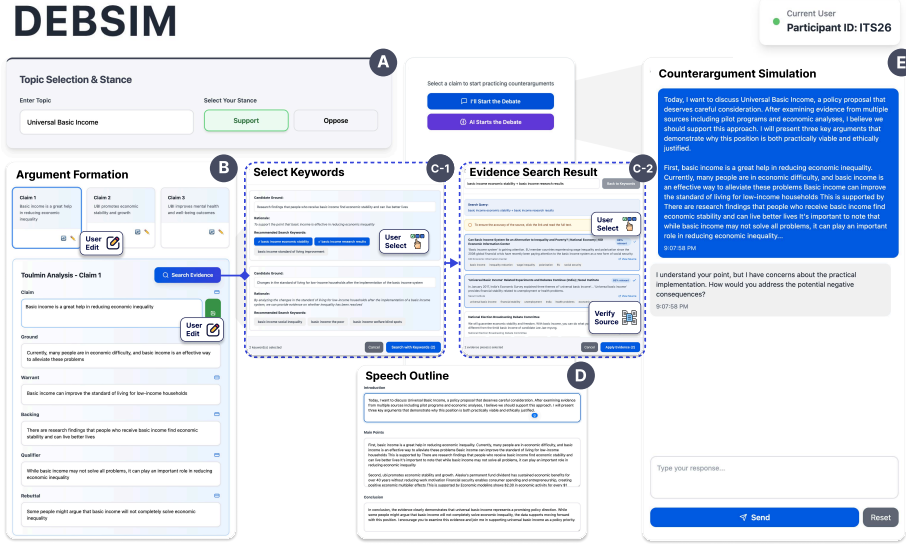


Fig. 1. The DEBSIM interface, with three friction points. (A) Topic Selection & Stance. (B) Argument Formation (Stage 1): editable Toulmin claim cards; the User Edit marker is the first friction point. (C-1, C-2) Keyword Selection and Evidence Search (Stage 2): sources are ranked but not auto-inserted, and the User Select and Verify Source markers form the second friction point. (D) Speech Outline: auto-generated script, continuously updated from the current KC structure. (E) Counterargument Simulation (Stage 3): learners type free-text responses to Devil’s Advocate challenges—the third friction point.

Stage 1: Argument Formation (Fig. 1A–B). The learner enters a topic and stance. DEBSIM generates Toulmin-structured suggestions as editable cards—one per KC. These are starting points: the learner must review, modify, and extend each card before proceeding. This editing requirement is the first friction point, encouraging active engagement with argument structure. Saved edits trigger reprocessing for logical coherence across KCs.

Stage 2: Evidence Enhancement (Fig. 1C). The learner selects search keywords from system-generated candidates, triggering searches via Google Search and Scholar APIs. Retrieved sources are ranked but not automatically inserted: the learner must explicitly select sources, assign each to a specific KC, and encounter a verification prompt encouraging engagement with the original source. This non-automatic insertion is the second friction point—the system handles search execution (mechanical) while evidence evaluation and integration remain with the learner (cognitive).

Stage 3: Counterargument Simulation (Fig. 1E). A dialogue interface presents AI-generated counterarguments targeting weaknesses in the learner’s Toulmin structure via a Devil’s Advocate approach [27]. The learner must type free-

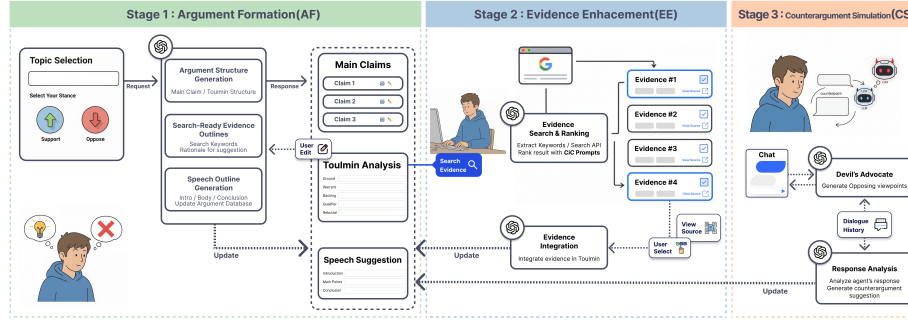


Fig. 2. DEBSIM’s three-stage backend workflow. Stage 1 (Argument Formation) generates Toulmin-structured claims and search-ready evidence outlines from the learner’s topic and stance input; Stage 2 (Evidence Enhancement) retrieves, ranks, and integrates user-selected sources via CiC prompts; Stage 3 (Counterargument Simulation) analyses learner responses and generates targeted counterarguments. Dashed arrows indicate user edit cycles; solid arrows indicate system processing flows.

text responses rather than selecting pre-generated options—a generation requirement [16] that is the third friction point.

Throughout all stages, a Speech Outline (Fig. 1D) is continuously updated from the current KC structure. Encountering a counterargument in Stage 3 may prompt a return to Stage 1 to restructure claims or to Stage 2 to find additional evidence, enabling iterative revision across the full workflow.

3.2 Backend Architecture and Pedagogical Model

Fig. 2 shows the backend workflow. Each stage runs an LLM pipeline that produces scaffolding content, but pedagogically meaningful decisions always require explicit learner action.

The domain model structures knowledge around six Toulmin KCs, each independently scaffoldable and assessable [13]. In Stage 1, an LLM generates initial KC suggestions; in Stage 2, a Corpus-in-Context (CiC) method [12] ranks retrieved evidence by relevance; in Stage 3, the LLM analyses the learner’s KC structure to identify weaknesses and generates targeted counterarguments. The student model tracks learner state through interaction logs—card edits (Stage 1), evidence selections and verification clicks (Stage 2), simulation turns (Stage 3)—monitoring KC completeness to inform counterargument targeting and revision prompting.

This architecture produces a two-level pedagogical loop. The inner loop (within-stage) cycles through scaffold, practice with friction, and feedback. The outer loop (across-stage) emerges from non-linear navigation: when Stage 3 counterarguments expose KC weaknesses, learners revisit earlier stages, revise, and return. Table 1 summarises friction mechanisms, targeted KCs, and theoretical bases.

Table 1. Productive friction mechanisms across DEBSIM’s three stages.

Stage	Friction	Learner Process	Target KCs	Theory
1. Formation	Editable KC cards; modification required	Argument structuring	Claim, Warrant	Scaffolding [28]
2. Evidence	Non-automatic insertion; source verification	Evidence evaluation	Data, Backing	Desirable difficulty [15]
3. Simulation	Typed responses to adversarial challenges	Counter argument generation	Qualifier, Rebuttal	Generation effect [16]

4 Evaluation

4.1 Study Design and Rationale

We conducted a within-subjects experiment with 20 university students (13 male, 7 female; $M_{\text{age}}=26.0$, $SD=2.1$; IRB-approved). Following VanLehn’s [14] framework distinguishing performance from learning, our evaluation assesses transfer: participants used DEBSIM only during preparation and had no system access during live debate.

Participants prepared arguments for two topics of comparable controversy and accessibility—Universal Basic Income (UBI) and Number of Parliament Members (NPM)—with one topic using DEBSIM and the other using self-directed preparation (web searches, notes, ChatGPT), an ecologically valid baseline reflecting how novices typically prepare today. Concretely, the 2×2 within-subjects design crossed topic (UBI / NPM) with condition (DEBSIM / self-directed), yielding four topic-condition orderings counterbalanced via Latin square with 5 participants per ordering (e.g., UBI-DEBSIM first then NPM-control). Live debates always used the same topic as the immediately preceding preparation session.

4.2 Procedure

Fig. 3 illustrates the study flow. Participants formed initial stances (6-point Likert [34]) and were paired with opposing viewpoints. Each participant completed both conditions in counterbalanced order: 20 min of preparation (preceded by a 10-min DEBSIM tutorial in the DEBSIM condition only), followed by a 15-min structured Zoom debate and a post-debate survey. Participants had no system access during debate; all performance reflects retained knowledge. Individual 30-minute exit interviews explored strategies and reflections.

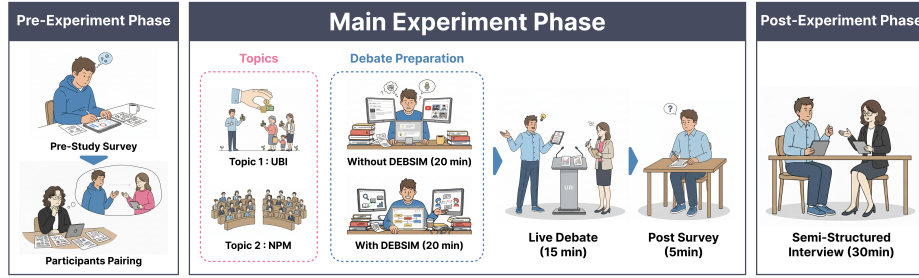


Fig. 3. Study procedure. Pre-Experiment Phase: pre-study survey and participant pairing. Main Experiment Phase: each participant prepared for both topics (UBI and NPM, 20 min each) under counterbalanced conditions, followed by live Zoom debate (15 min) and post-survey (5 min); no system access was allowed during debate. Post-Experiment Phase: semi-structured interview (30 min). Topic-condition-order assignment followed a 2×2 Latin square (4 orderings, $n=5$ per ordering).

4.3 Measures

Interaction logs recorded discrete DEBSIM actions with timestamps, serving as engagement proxies and capturing stage transitions. Transcript coding ($\kappa=.82$, two independent coders) applied Toulmin structural coding to opening arguments (six KCs) as the primary transfer metric, and Walton functional coding (challenge, clarification, confirmation, concession) to interactive exchanges. Raw counts were normalised by total coded instances. Pre/post surveys measured stance, familiarity, and perceived quality (self-evaluation: 7 items [32]; opponent evaluation: 4 items [33]; overall quality) on 7-point Likert scales. Table 2 lists all survey items; abbreviated labels denote: Supporting Evidence (Supp. Evidence), Source Validation (Source Valid.), Persuasive Delivery (Pers. Delivery), Easy Comprehension (Easy Compr.), Effective Rebuttal (Eff. Rebuttal), Opponent Understanding (Opp. Underst.), Evidence Assessment (Evid. Assess.), Perspective Taking (Persp. Taking), and Position Reflection (Position Refl.). Qualitative analysis of interviews used inductive and deductive coding (611 initial codes, four researchers, three final themes).

Quantitative comparisons used paired Wilcoxon signed-rank tests with Benjamini-Hochberg correction (FDR $q < .05$). KC effect sizes (Cohen’s d) serve as learning gain measures [14]. Exploratory Spearman correlations between stage interactions and outcomes are reported as hypothesis-generating evidence.

5 Findings

5.1 Engagement: Iterative Inner- and Outer-Loop Behaviour

Interaction logs revealed iterative rather than linear workflows (Fig. 4). Stage 2 (Evidence Enhancement) received the most visits (27 total transitions), with

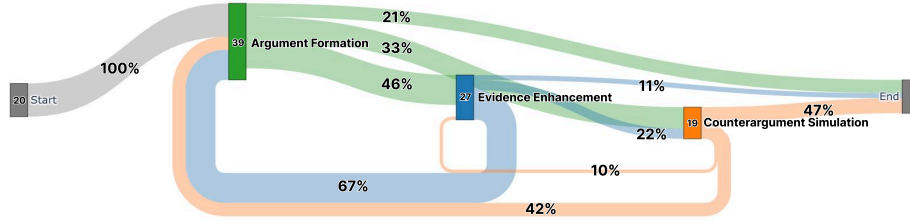


Fig. 4. Aggregated stage-transition patterns across all participants ($N=20$). Node values indicate total interaction counts (Argument Formation: 39, Evidence Enhancement: 27, Counterargument Simulation: 19).

67% of transitions from Stage 2 cycling back to Stage 1, forming a tight inner loop between KC structuring and evidence integration. Stage 1 (Argument Formation) received 39 transitions, with participants actively adapting scaffolded KC structures rather than accepting defaults. Stage 3 (Counterargument Simulation) received 19 transitions; roughly half (52%) of post-simulation transitions returned to earlier stages—illustrating outer-loop behaviour where adversarial feedback prompted revision of previously constructed arguments.

P19 illustrated friction-driven selectivity: “some suggestions included blog posts, which didn’t feel reliable”—showing critical evaluation rather than wholesale acceptance.

5.2 Learning Transfer: KC Completeness and Functional Changes

We analysed 20 debate transcripts (599 coded utterances). All data reflect unscripted performance without system access.

KC Completeness. Fig. 5 summarises the distribution of Toulmin components in debate openings across conditions. The total number of KC instances per opening was higher in the DEBSIM condition ($M=9.9$, $SD=2.8$) than in the control condition ($M=6.1$, $SD=2.3$). The difference was distributed unevenly across KCs, with backing and qualifier showing the most pronounced patterns. In the control condition, backing was largely absent ($M=0.4$, $SD=0.6$), whereas DEBSIM participants incorporated backing more consistently ($M=1.5$, $SD=0.8$). Qualifier usage followed a similar pattern ($M=0.6$, $SD=0.5$ vs. $M=1.1$, $SD=0.5$), consistent with Stage 3’s counterargument simulation encouraging anticipatory hedging. Claim ($M=2.1$ vs. $M=2.9$), data ($M=1.6$ vs. $M=2.4$), and warrant ($M=1.4$ vs. $M=2.0$) also showed higher values in the DEBSIM condition. These patterns suggest that DEBSIM scaffolding was associated with more structurally complete argumentation in unassisted settings, with the largest differences appearing in components typically underrepresented in novice arguments.

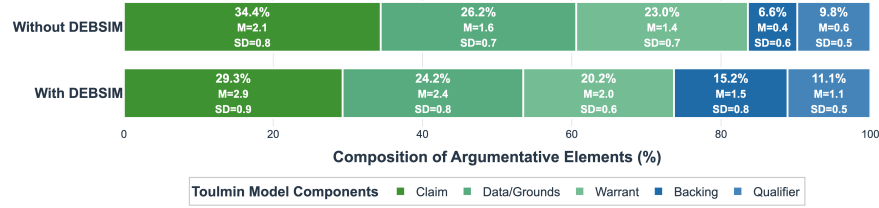


Fig. 5. Composition of Toulmin knowledge components in debate opening arguments across conditions ($N=20$ per condition). Segments show percentage share, mean count (M), and standard deviation (SD) per opening.

Functional Changes. Beyond structural completeness, the quality of interactive exchanges differed between conditions. Challenge frequency remained comparable, but the nature of challenges shifted: DEBSIM-condition challenges tended toward evidence-based inquiry rather than adversarial negation. Clarification requests became more source-focused (e.g., “Could you confirm the source of that statistic?”). The most notable difference appeared in concessions, which were rare in the control condition ($n=3$) and more frequent in the DEBSIM condition ($n=14$). Concessions in the DEBSIM condition tended to be strategic rather than defensive (e.g., “I concede employment effects are uncertain. However, I maintain...”). These functional changes are noteworthy because they emerged in unscripted exchanges structurally different from the preparation tool, suggesting that the patterns were not simply reproductions of scaffolded content.

Exploratory Correlations. Stage 3 interactions showed a moderate correlation with concession frequency ($r_s=.41$, $p=.07$); Stage 2 interactions with backing density ($r_s=.38$, $p=.10$). Neither reached significance at $\alpha=.05$, and the small sample ($N=20$) limits interpretability. We report these as hypothesis-generating observations: the stage-specific pattern (Stage 2 $\rightarrow$ backing, Stage 3 $\rightarrow$ concession) is consistent with the intended pedagogical functions and may warrant investigation through ablation studies with larger samples.

5.3 Self-Reported Outcomes

Table 2 presents post-survey results. Using paired Wilcoxon signed-rank tests with Benjamini–Hochberg correction (FDR $q<.05$), five of twelve measures reached significance. Among self-evaluation items, participants rated Claim Clarity ($W=29.5$, $p=.025$) and Effective Rebuttal ($W=30.5$, $p=.027$) higher in the DEBSIM condition. Among opponent evaluation items, Opponent Understanding ($W=29.0$, $p=.039$) and Perspective Taking ($W=22.0$, $p=.028$) differed between conditions. Overall Debate Quality showed the largest difference ($W=15.5$, $p=.001$).

Table 2. Post-survey results ($N=20$): Wilcoxon signed-rank tests. * $p<.05$, ** $p<.01$ after Benjamini–Hochberg correction (FDR $q<.05$).

Category	Measure	w/ DEBSIM		w/o DEBSIM		W	p
		Md(IQR)	M(SD)	Md(IQR)	M(SD)		
Self Eval.	Claim Clarity	6.00(1.00)	5.50(1.02)	5.00(1.25)	4.35(1.53)	29.5	.025 *
	Supp. Evidence	6.00(2.00)	5.15(1.28)	5.00(3.00)	4.70(1.73)	60.0	.430
	Source Valid.	5.00(2.00)	4.85(1.19)	4.00(3.25)	4.25(2.02)	69.5	.301
	Pers. Delivery	5.00(1.25)	5.05(1.12)	5.00(1.00)	4.95(1.77)	65.0	.875
	Easy Compr.	5.50(2.00)	5.20(1.08)	5.00(2.25)	4.65(1.62)	43.0	.183
	Logical Flow	6.00(1.25)	5.50(0.97)	5.00(3.00)	4.60(1.71)	36.0	.086
	Eff. Rebuttal	6.00(2.00)	5.15(1.28)	3.50(2.00)	3.85(1.56)	30.5	.027 *
Opp. Eval.	Opp. Underst.	6.00(1.00)	6.00(0.89)	5.00(2.25)	5.15(1.53)	29.0	.039 *
	Evid. Assess.	6.00(2.00)	5.30(1.31)	5.00(2.25)	4.70(1.58)	56.0	.187
	Persp. Taking	6.50(1.00)	6.30(0.95)	5.00(2.00)	5.30(1.62)	22.0	.028 *
	Position Refl.	3.50(2.25)	3.75(1.76)	3.00(2.25)	3.15(1.56)	64.0	.120
Overall Eval.	Debate Quality	6.00(0.25)	5.90(0.99)	5.00(2.00)	4.20(1.25)	15.5	.001 **

The measures that survived correction correspond to dimensions where behavioural differences were observed in transcript coding (argumentation structure, rebuttal quality, and perspective-taking), while measures related to delivery and general comprehension did not differ. This alignment between self-report and behavioural data suggests that reported differences reflect perceived competence tied to specific argumentation skills rather than a generalised halo effect.

5.4 Learner Reflections: Transfer Mechanisms

Three interview themes illuminated transfer mechanisms. First, active integration: participants described deliberate engagement rather than passive adoption (P18: “I didn’t just use the data as-is—I checked it and supplemented it with my own searches”). Second, friction catalysing ownership: AI limitations enhanced engagement (P18: “Questioning the system’s reliability forced me to research more, making my arguments feel more ‘mine’”), illustrating how imperfect outputs within verification scaffolds may function as generative features. We characterise this sense of personal investment—where learners treat arguments as their own rather than system-generated products—as cognitive ownership, a perceived state that participants linked to friction-induced effortful processing. Third, counterargument simulation enabling adaptation: P10 reported that adversarial rehearsal supported adaptation during live debate. Participants also raised over-reliance concerns (P12, P5), underscoring the need for adaptive scaffold fading.

6 Discussion

6.1 Addressing the Assistance Dilemma Through Productive Friction

Our findings suggest that structured scaffolding with productive friction during preparation was associated with observable differences in live debate con-

ducted without system access. A recent systematic review of 88 empirical studies on LLMs in education identifies intelligent tutoring as the dominant application but flags over-reliance as a pervasive challenge [40]. DEBSIM’s selective-automation strategy offers one design-level response: providing substantial AI assistance while preserving effortful processing at cognitively meaningful decision points [37]. The proposed mechanism is the separation of mechanical and cognitive automation: the system provides support for tasks such as search execution and formatting while requiring that cognitively meaningful decisions remain with the learner. Several patterns support engagement beyond memorisation: iterative outer-loop cycling, qualitative shifts in discourse functions, and participants’ accounts of active adaptation [15, 16]. However, the within-subjects design and sample size warrant caution in interpreting these associations.

Although our design cannot isolate individual stage effects, patterns align with stage-specific contributions: KC completeness differences corresponded to Stage 1’s structuring scaffolds; evidence citation patterns to Stage 2’s verification friction; and concession patterns to Stage 3’s counterargument simulation.

6.2 Design Implications

Our findings point toward four tentative principles for argumentation tutoring. First, systems may benefit from scaffolding the process of argument development rather than generating the product, with assessment conducted after scaffold removal [14]. When systems produce polished outputs on demand, learners risk engaging in surface-level editing rather than generative reasoning; designing for process scaffolding redirects cognitive effort toward the decisions that matter. Second, the assistance dilemma may be partially addressed through selective automation—automating mechanical operations while preserving effort at cognitive decision points [37]. Our evidence suggests that evidence evaluation and counterargument generation are cognitive operations where learner effort yields meaningful returns. Third, counterargument simulation that bridges monological and dialogical reasoning may help learners apply Toulmin KC construction skills to interactive exchanges. Fourth, the substantial back-cycling observed in stage transitions (Fig. 4) suggests that non-linear, iteratively revisitable workflows may benefit argumentation tutoring by fostering recursive refinement that strengthens argument coherence.

6.3 Limitations and Future Work

We frame this study as a pilot establishing the feasibility of transfer-based evaluation, and three limitations constrain its interpretation. First, the study is underpowered for detecting fine-grained effects and short in duration: $N=20$ limits generalisability, and the 20-minute session with immediate post-test cannot speak to durable learning. Scaling to larger samples and adding delayed post-tests across multiple sessions would help distinguish durable learning from short-term priming [14]. Second, the self-directed control—though ecologically valid—does not isolate which aspect of DEBSIM drives the observed differences.

Stage-wise ablations (non-editable Stage 1 cards, auto-inserted Stage 2 evidence, or multiple-choice Stage 3 responses) against a structured-ChatGPT baseline are a planned next step; a between-subjects replication would additionally address within-subjects carry-over that Latin-square counterbalancing can only partially mitigate. Third, DEBSIM’s friction is fixed rather than adaptive, and its binary scaffold transition from full support to full removal may be suboptimal compared with gradual fading [14, 37]. Integrating KC mastery estimation for adaptive friction calibration, together with cross-session scaffold fading, is a natural direction for aligning DEBSIM with the agentic-tutoring paradigm.

7 Conclusion

This study assessed whether LLM-based argumentation scaffolding transfers to unassisted performance. DEBSIM embeds productive friction across three preparation stages—argument structuring, evidence validation, and counterargument simulation—automating mechanical operations while preserving learner effort at cognitive decision points. In a within-subjects experiment, preparation with DEBSIM was associated with higher Toulmin KC completeness, evidence-based discourse, and strategic concessions in unassisted live debate. This work contributes a transfer-based evaluation methodology distinguishing learning from supported performance, productive friction as a design strategy for the assistance dilemma in the LLM era, and an inner/outer-loop KC scaffolding architecture supporting argumentation transfer.

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